

RICHA KAPLESH

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EDUCATION

B.Tech — Artificial Intelligence & Machine Learning
CGC Landran

CGPA: 8.5
Expected 2026

EXPERIENCE

Student Intern — Nokia

August 2025 – Present

- Built production analytics dashboards in **React + TypeScript** using AG Grid; implemented custom Apache Superset plugins and microfrontends for internal data visualization
- Mapped and transformed complex data structures using **JSONata**, integrated REST APIs, and wrote PostgreSQL queries for backend data workflows
- Worked in a **Docker + Ubuntu/WSL** environment; contributed to Python tooling and internal data pipeline scripts

PROJECTS

QueryMind v2 — Python · FastAPI · LangChain · FAISS · BM25 · Groq · React · TypeScript

- Built an agentic document QA system with **query-intent-aware routing** — PDF queries use hybrid BM25 + FAISS retrieval with cross-encoder reranking; CSV queries trigger a scipy-based statistical analysis pipeline
- Implemented a **tool-calling generator using Groq function calling** to orchestrate document search, CSV statistical analysis, and web search tools — each as an independently callable tool
- Designed with **SOLID principles**: extraction, chunking, embedding, retrieval, prompt building, and tool execution are fully decoupled single-responsibility modules
- Added **OCR fallback via Tesseract** for scanned PDFs, SSE-based streaming responses, and in-session conversation history

Fake News Detector — Python · scikit-learn · FastAPI · React · TypeScript

- Ran **9 controlled ML experiments** comparing TF-IDF vs Word2Vec features, logistic regression vs random forest classifiers, and source credibility feature engineering — achieving 88% accuracy on the final configuration
- Engineered features including **source credibility scores and linguistic pattern signals** on top of baseline bag-of-words to isolate the contribution of each feature set across experiments
- Built with a **FastAPI backend + React + TypeScript frontend**; structured as a reproducible experiment suite with documented hypothesis, feature set, and accuracy delta per run

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript, R
- **AI & ML:** RAG, LangGraph, Multi-Agent Systems, LLM Integration, Tool Calling / Function Calling, Vector Databases, RAGAS Eval, End-to-End ML Pipelines
- **Backend & APIs:** FastAPI, REST APIs, PostgreSQL, SQL
- **Frontend:** React, TypeScript, CSS, HTML
- **Infrastructure:** Docker, WSL, Ubuntu/Linux
- **Data & Analytics:** Apache Superset, JSONata, Superset Plugins